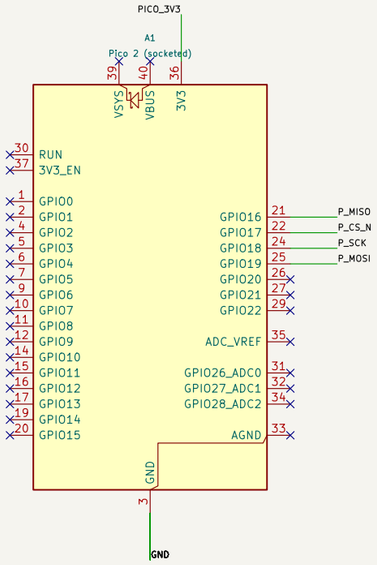
SPU SOUTHBRIDGE REV 1 — TRANSLATOR CANDIDATE

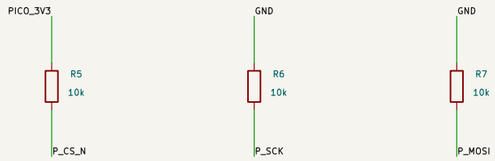
DRAFT / NOT FOR FABRICATION — independent 3.3 V domains; dedicated SPI mode 0

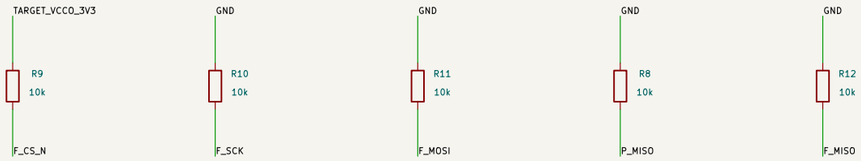


Pico USB powered; VBUS / VSYS / 3V3_EN / RUN unused on carrier.

A1 footprint: two 1x20 sockets; verify fit before layout.

Idle bias — each net uses its own domain only





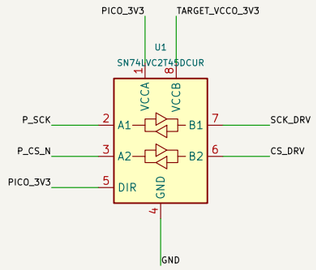
Power-off isolation candidate; no UVLO / power-good / OE.

Brownout transaction blocking is NOT implemented.

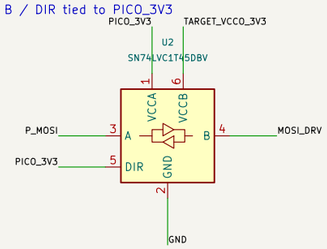
No UART / flash / other cross-board bypass paths included.

No bench qualification; VCCO identity and ramp tests outstanding.

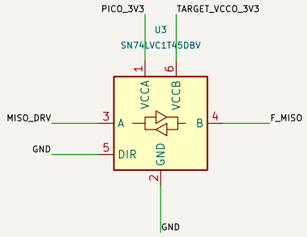
U1: A -> B / DIR tied to PICO_3V3

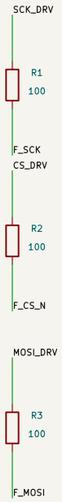


U2: A -> B / DIR tied to PICO_3V3



U3: B -> A / DIR tied to GND



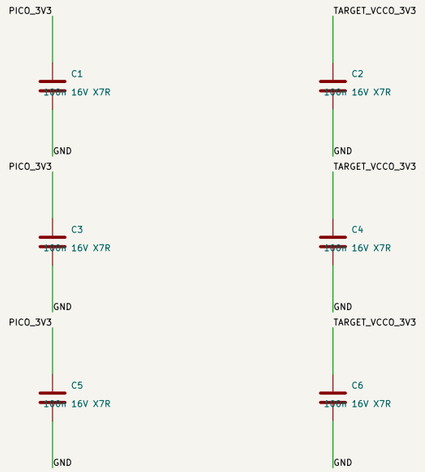


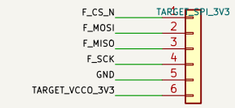


R1–R4: place at translator outputs. 100 ohm is a starting value.

Bypass pairs: C1/C2 U1; C3/C4 U2; C5/C6 U3.

Place at IC supply pins; short ground returns.







No rail tie. Common ground must connect first.

TARGET rail powers VCCB; not sense-only.

J2 cable to THIS Wukong remap:

1 CS# -> J11-7 2 MOSI -> J11-9

3 MISO -> J11-10 4 SCK -> J11-8

5 GND -> J11-11 6 VCCO -> J11-12*

* HOLD: verify J11-12 equals actual bank rail.

No 1:1 cable to Wukong or Tang.

Rev 1 power: USB -> Pico regulator; no extra regulator.

Pico external 3V3 budget: <=100mA initial target.

Planning slots (mA), NOT measured:

Interface 10 | INA226 5 | I2C sensors 10

Storage 50 | LEDs 2 | Margin 23 = 100

Modules TBD; check peak/startup current before fitting.

Target VCCO: B-sides only; never peripheral power.

NOT FOR FABRICATION; no brownout threshold

Sheet: /
File: southbridge_rev1_candidate.kicad_sch

Title: SPU Southbridge Rev 1 — translator candidate

Size: A3	Date: 2026-11-05	**Rev: R1-CANDIDATE**
KiCad E.D.A. 10.0.7		Id: 1/1